

Machine Learning 1

LAB EXERCISE 3

This lab exercise must be submitted by May 16nd, 2025 at 10:00 am.
Late submissions will not be accepted and will be marked as zero.

This lab exercise is mandatory and could be assessed.
It is worth 1% of your total final grade for this course.

Submission Instructions

Submit your files through EClass. The files you submit cannot be read by any other students. You can replace your submission as many times as you like by resubmitting it, although only the last version sent is kept.

If you have last-minute problems with your EClass submission, email your assignment as an attachment to alberto.paccanaro@fgv.br with the subject "URGENT – LAB 3 SUBMISSION". In the body of the message, explain the reason for not submitting it through EClass.

All the work you submit should be solely your own work. Coursework submissions will be checked for this.
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In this lab, you will implement Linear Regression and Logistic Regression models. For linear regression, you will implement both the closed form solution as well as the stochastic gradient descent optimization strategy. For logistic regression, you will implement both the stochastic gradient descent as well as the Iterative Reweighted Least Squares algorithm.

DATASET DESCRIPTION

For Linear Regression, you will be using the **housing dataset** available on the Eclass page. This dataset is constituted by 506 points in 14 dimensions. Each point represents a house in the Boston area, and the 14 attributes that you find orderly in each column are the following:

- CRIM - per capita crime rate by town
- ZN - proportion of residential land zoned for lots over 25,000 sq.ft.
- INDUS - proportion of non-retail business acres per town.
- CHAS - Charles River dummy variable (1 if tract bounds river; 0 otherwise)
- NOX - nitric oxides concentration (parts per 10 million)
- RM - average number of rooms per dwelling
- AGE - proportion of owner-occupied units built prior to 1940
- DIS - weighted distances to five Boston employment centres
- RAD - index of accessibility to radial highways
- TAX - full-value property-tax rate per \$10,000
- PTRATIO - pupil-teacher ratio by town
- B - $1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town
- LSTAT - % lower status of the population
- MEDV - Median value of owner-occupied homes in \$1000's

This dataset is normally associated with 2 regression tasks: predicting NOX (the nitrous oxide level s); and predicting MEDV (the median value of a home).

For Logistic Regression, you will be using the **occupancy dataset** that you will find on the Eclass page. This dataset contains 20560 points in 5 dimensions. Each point represents a room described by 5 attributes (first 5 columns). Each room belongs to one of 2 possible classes (last column). For your convenience the dataset has already been divided into a training set, a validation set and a test set. The description of the attributes is as follows:

- Temperature, in Celsius
- Relative Humidity in percentage
- Light, in Lux
- CO2, in ppm
- Humidity Ratio, in kgwater-vapor/kg-air

The classes to predict are 0/1 denoting the room occupancy, that is: was a person in there or not? (Occupancy detection in buildings has been estimated to save energy in the order of 30 to 42%, when used with control systems).

YOUR TASKS

A) Build a Linear Regression model. You will need to implement it twice:

- i. using the Normal Equations seen in class.
- ii. using stochastic gradient descent.

TIP 1: when you implement the stochastic gradient version it will be crucial to plot the Sum of Squares error as a function of the iteration.

TIP 2: you will need to initialize the weights. Start by choosing them randomly, gaussian distributed, with a very small variance...

Run some experiments with your stochastic gradient descent version.

Think about these questions:

- when using stochastic gradient descent, what happens with different values of the step parameter? Does the system always converge to the same solution?
- with the stochastic gradient descent, are you getting the same solution that you get with the closed form implementation?
- What happens if you initialize your weights to values that are too big?

B) Build a Logistic Regression model. You will need to implement it twice:

- i. using the stochastic gradient descent.
- ii. using Iterative Reweighted Least Squares algorithm.

TIP 1: it will be interesting to plot the Cross Entropy error vs the iteration.

TIP 2: you will need to initialize the weights. Start by choosing them randomly, gaussian distributed, with a very small variance.

Run some experiments and think about these questions:

- Do the two implementations always converge to the same solution?
- What happens if the step size of your stochastic gradient descent is too large?
- What happens if you initialize your weights to values that are too large?

Have fun ! 😊